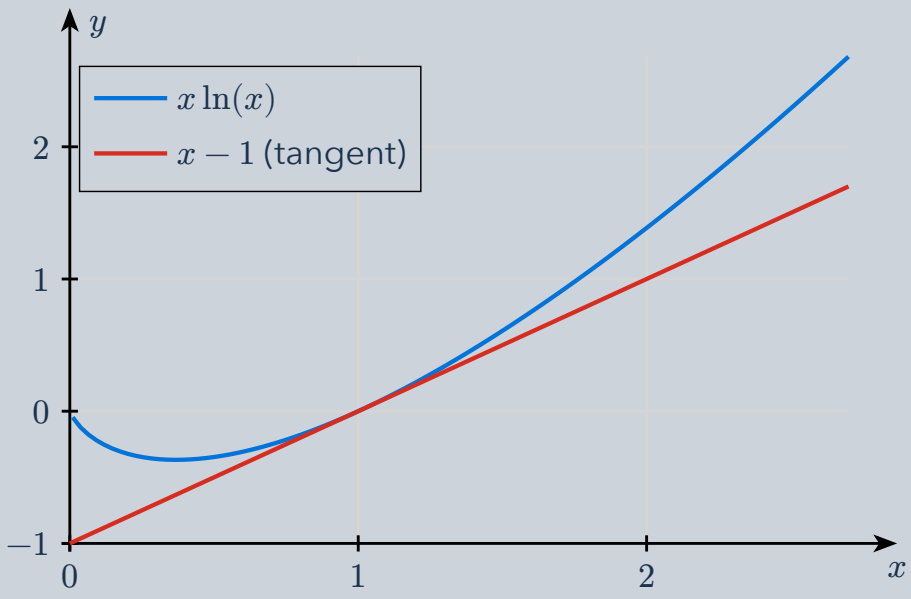
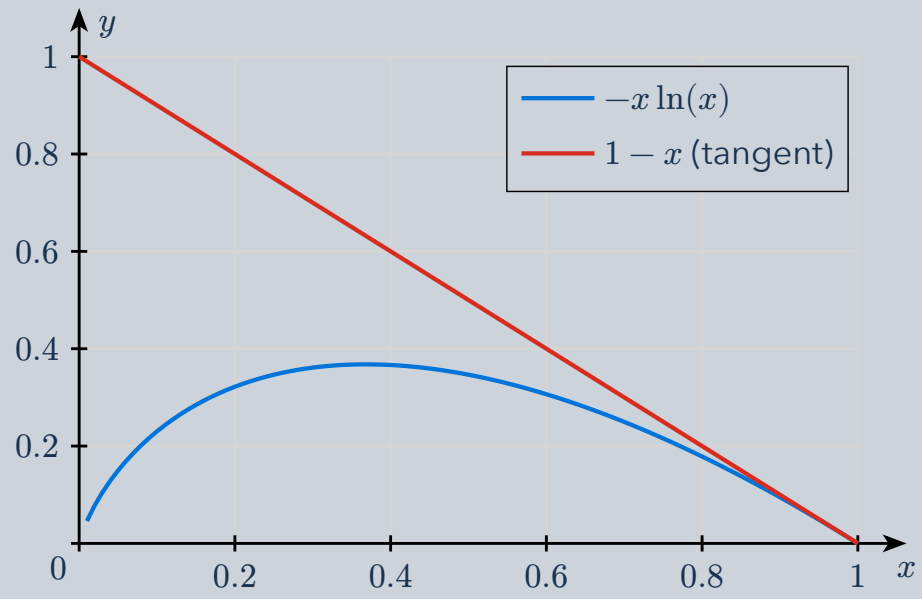
Curvature tells us how a function compares with its tangents, chords, and averages. Convex bends upward; concave bends downward.

1 Convex: above a tangent



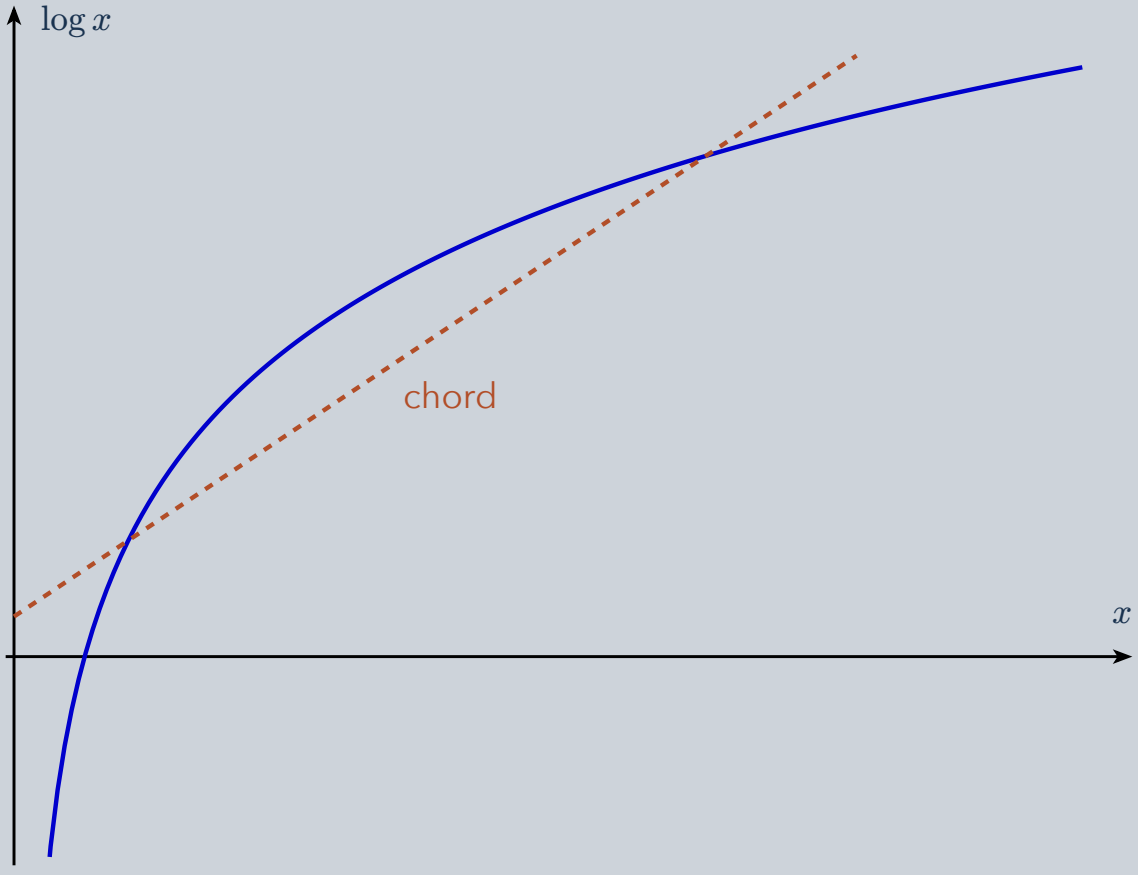
For $f(x) = x \ln x$ $(x > 0)$, the tangent at $x = 1$ is $x - 1$. Thus $x \ln x \geq x - 1$, with equality at the contact point.

2 Concave: below a tangent



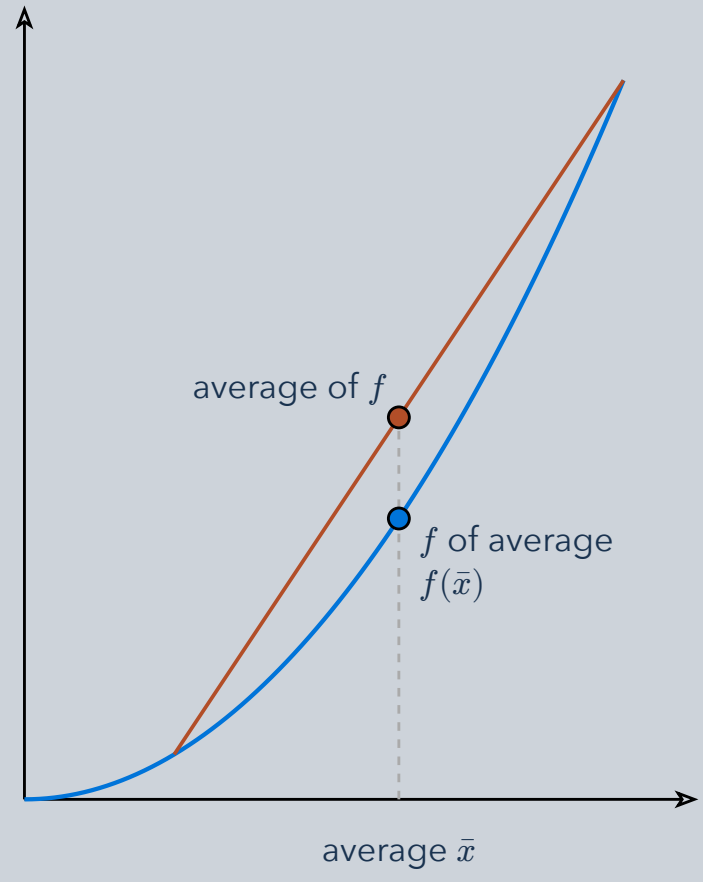
Negating $x \ln x$ flips its curvature: the concave $-x \ln x$ lies below its tangent $1 - x$ at $x = 1$. A straight line is both convex and concave.

3 Concave: above a chord



Between the two points where the dashed chord meets it, the concave logarithm lies above the chord. Jensen's inequality extends this from two points to any weighted average.

4 Average first, or apply f first?



For convex f, $f\left(\sum_i w_i x_i\right) \leq \sum_i w_i f(x_i)$ when $w_i \geq 0$ and $\sum_i w_i = 1$. For concave f, reverse the sign.

Tangent = local slope; chord = interpolation between two points. Jensen compares the function of an average with the average of the function.